EXL CAMPUS HIRE PROGRAMME

Capstone Project Presentation

StockSense AI
Evidence-Based Stock Decision Support

A decision-support platform that combines forecasts, market news, risk signals and grounded reasoning.

Track / Batch [Add track · batch]

Presentation Date 07 Aug 2026

TEAM (MAX 3)

- [Add team member names]

2 Problem & Business Context

Stock analysis requires more than a single price prediction.

INDUSTRY & FUNCTION

- Financial Services — Investment Research
  Stock analysis & decision support
  Primary user: analyst / investor

THE PROBLEM

- • Fragmented prices, indicators, fundamentals, macro signals and news
  • Raw predictions can be black-box and hard to trust
  • Earnings and guidance can quickly change the picture
  • AI can fuse quantitative and qualitative evidence

CURRENT-STATE COST

Source build does not provide measured baseline costs.
Manual effort: Not measured
Turnaround time: Not measured
Error / rework rate: Not measured
Cost / leakage: Not measured

3 Solution Overview

A multi-stage workflow turns one ticker into an evidence-backed analysis.

WHAT THE SOLUTION DOES

- • Forecast next-day returns using four complementary models
  • Retrieve, deduplicate, rank and score relevant market news
  • Fuse forecasts, sentiment, fundamentals and macro context

Success criteria: grounded recommendations with confidence, risks and sources.

WHO USES IT & HOW

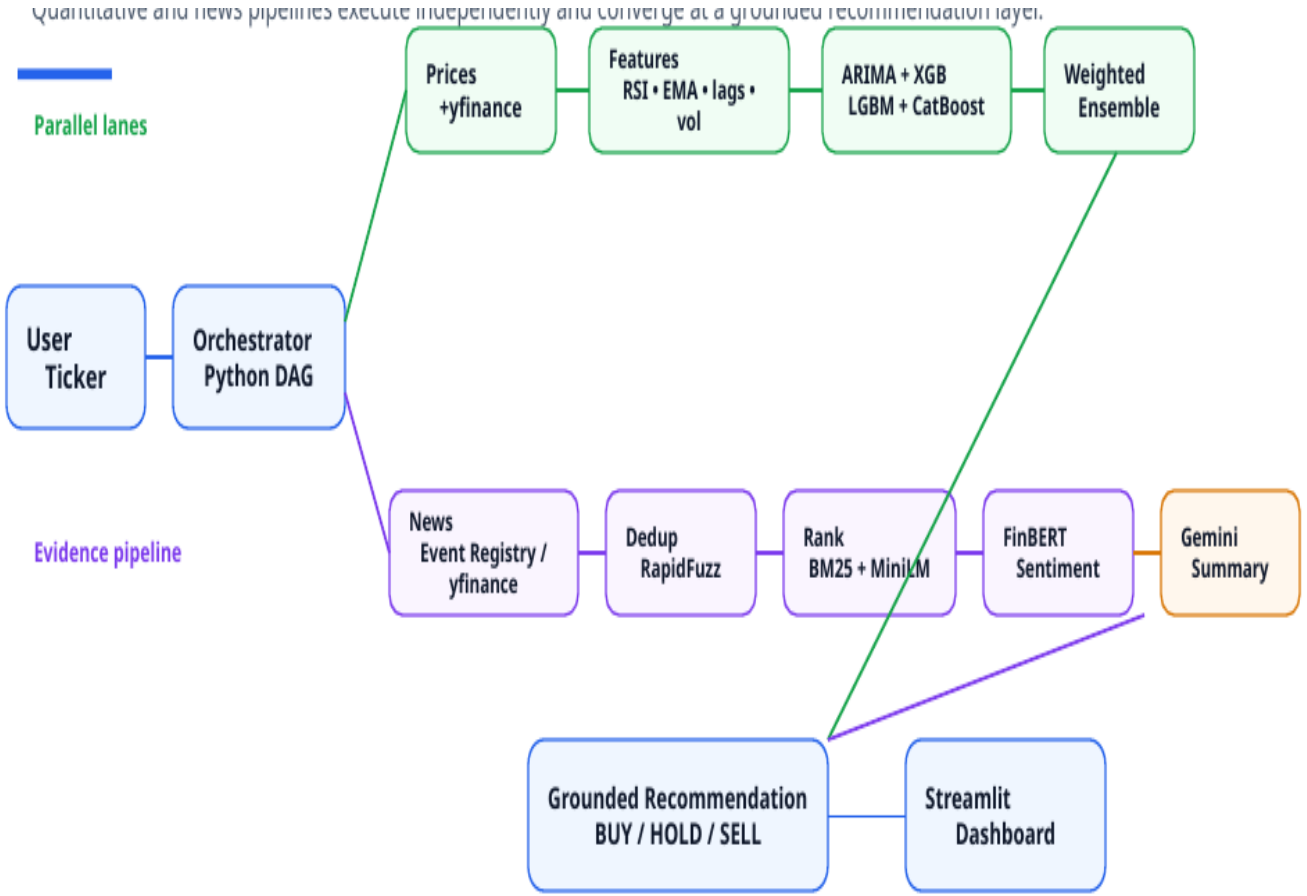
- Primary user: analyst / investor
  Workflow: enter ticker → review forecast, news, risk → inspect recommendation

SCOPE

- In scope: forecasting, retrieval, sentiment, fusion, dashboard, history, screener and comparison
  Out of scope: financial advice and guaranteed market prediction

4 Architecture & Tech Stack

Independent quantitative and evidence pipelines converge at grounded recommendation.



TECH STACK

- *[Models · frameworks · orchestration]*
- *[Data stores · deployment surface]*

Python DAG • yfinance • statsmodels
XGBoost • LightGBM • CatBoost • scikit-learn
BM25 • MiniLM • RapidFuzz • FinBERT
Gemini • LangChain • Streamlit • Plotly
SQLite / SQLAlchemy

KEY DESIGN DECISIONS

- *[Decision 1 — and why]*
- *[Decision 2 — and why]*

1. Separate quant and evidence pipelines to reduce noise and isolate failures.
2. Ground LLM reasoning in computed signals and retrieved articles, with deterministic fallback.

5 Solution in Action

Real StockSense AI screens from the working build.



SCREENSHOT 2

Caption: Technical analysis provides interpretable market context.

Technical dashboard: candlesticks, SMA/EMA, Bollinger Bands and RSI.



A real input going in, and the step that processes it

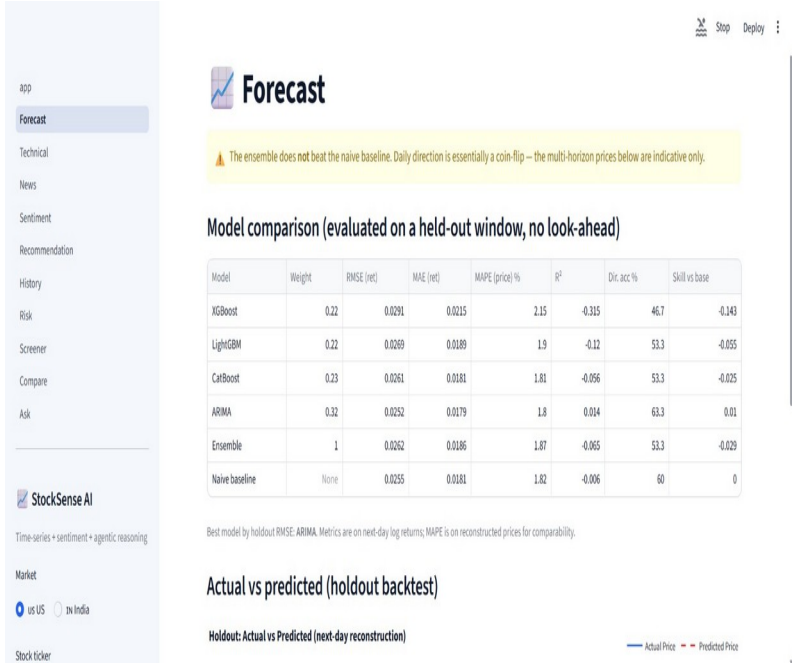
Sentiment dashboard: credibility-weighted FinBERT analysis across retrieved news.

Caption: News is reduced to relevant, scored evidence.

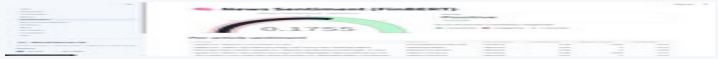
Real AAPL inputs shown from the supplied project presentation.

6 Outputs & Evidence

The working build exposes both model performance and uncertainty.



Forecast output: ensemble benchmarked against a naive baseline.



One edge case handled

SCREENSHOT 4

Edge evidence: credibility-weighted sentiment across 8 articles.

- *[Accuracy]*
- *[Latency]*
- *[Cost per run]*
- *[Success rate]*
- *[Evaluation method used]*

Accuracy: ensemble directional accuracy 53.3%
Naive baseline: 60%
Best displayed model: ARIMA, 63.3% directional accuracy
Evaluation: held-out window; no look-ahead

- *[Slide 2 number] → [number after the solution]*

Baseline directional accuracy 60% → Ensemble 53.3%

- *Fully working: […] · Mocked: […]*

Fully working: forecasting, retrieval, sentiment, recommendation, dashboard, history and analytics
Mocked: not identified in the supplied source presentation

7 Business Value & Scale-Up Path

Value claims are kept evidence-based; the source does not quantify time or cost savings.

Not measured	Not measured	53.3%
Hours saved per week	Cost or effort reduced	Ensemble directional accuracy

BASIS OF ESTIMATE

- Basis: supplied AAPL holdout results.
  The source provides model metrics, but no manual-effort baseline.
  Therefore time/cost ROI is not estimated.

PATH TO ENTERPRISE GRADE

- Security & access control
  Data governance and API-key management
  Integration, monitoring and scale
  Compliance / auditability of recommendations

KNOWN GAPS

- Daily directional skill is weak versus naive baseline.
  External APIs and market/news freshness require monitoring.
  Next: larger backtests, stronger validation and production controls.

Thank You

Questions & Discussion

Team [Add team members]

Track / Batch [Add track · batch]